27 STELLAR EVOLUTION

The Universe is mainly empty space within which are scattered large numbers of galaxies. Astronomers (scientists who study space and the Universe) believe that there are billions of galaxies in the Universe. The distances between galaxies are millions of times greater than the distances between stars within a galaxy. The distances between the stars in a galaxy are millions of times greater than the distances between planets and the Sun.



▲ Figure 27.1 Photograph of galaxies of the Universe taken by the Hubble space telescope

LEARNING OBJECTIVES

- Know that the Sun is a star at the centre of our Solar System
- Know that the Universe is a large collection of billions of galaxies; galaxies are a large collection of billions of stars; our Solar System is in the Milky Way galaxy
- Understand how stars can be classified according to their colour
- Know that a star's colour is related to its surface temperature
- Describe the evolution of stars of similar mass to the Sun through the following stages:
  - nebula
  - star (main sequence)
  - red giant
  - white dwarf
- Describe the evolution of stars with a mass larger than the Sun

PHYSICS ONLY

- Understand how the brightness of a star at a standard distance can be represented using absolute magnitude
- Draw the main components (parts) of the Hertzsprung–Russell diagram (HR diagram)

THE MILKY WAY

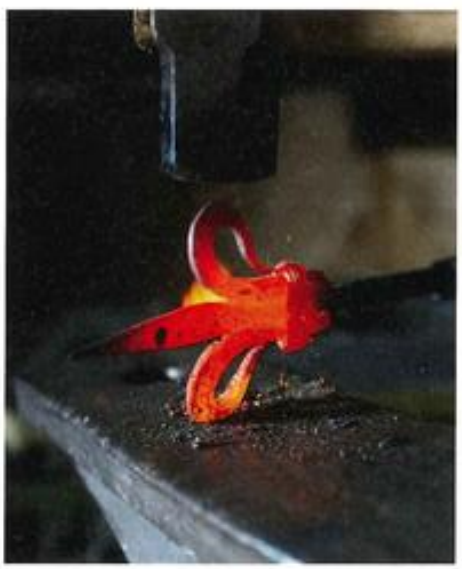
Our nearest star is the Sun. It is approximately 150 million kilometres from the Earth. Its surface temperature is approximately 6000 °C and temperatures within its core are about 15 000 000 °C. Attractive gravitational forces between stars cause them to group together in enormous groups called galaxies. Galaxies consist of billions of stars. Our galaxy is a **spiral galaxy** called the Milky Way. We are approximately two-thirds of the way out from the centre of our galaxy along one of the arms of the spiral.



▲ Figure 27.2 Our galaxy, the Milky Way



▲ Figure 27.3 Our galaxy takes the shape of a spiral, like the one shown here.



▲ Figure 27.4 The colour of the light emitted by this piece of iron tells us how hot it is.

CLASSIFYING STARS

Even within our galaxy we can see a wide variety of stars (Figure 27.2).

Looking up into a clear night sky, especially if we use a telescope, we can see that stars are not all identical. They have different colours, different levels of brightness and appear to be different sizes. Scientists who study the stars in detail have created classes or star groups based upon these similarities and differences.

Why are some stars white, some red and others yellow? Surprisingly we can find the answer to this question in a traditional blacksmiths. To shape a piece of iron without it breaking the iron must be very hot. A skilled blacksmith will know when the temperature of the iron is high enough simply by taking it out of the furnace (very hot oven) and looking at its colour. If it is glowing white or bright orange it is very hot. If it is a dull red it is much cooler and not ready to be shaped.

In a similar way the colours of stars tell us about their temperatures. A very hot star emits more blue in its spectrum and therefore looks blue, a medium star like our Sun looks yellow and cooler stars appear red.

EXTENSION WORK

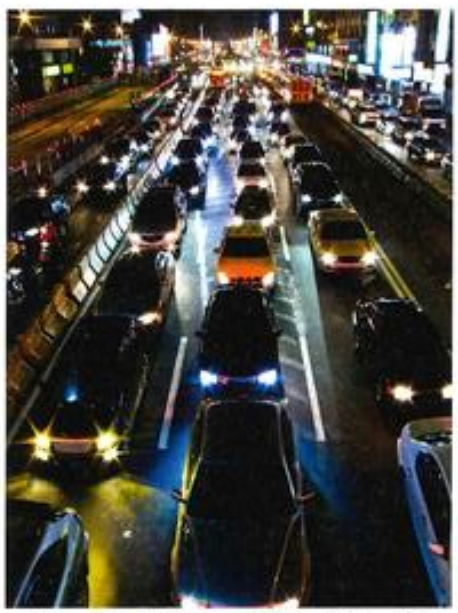
Scientists often subdivide these groups into 10 smaller groups by giving them a number between 0 and 9. Our Sun is classified as G2 – that is, it is two-tenths along from a G star towards a K star. (You will not be tested on this in your exam.)

Using the colours of stars and their surface temperatures, scientists have created seven different groups of stars. These groups are called O, B, A, F, G, K, M. O and B are the hottest stars and K and M the coolest.

Star classification	Surface temperature/K	Colour
O	more than 33 000	blue
B	33 000–10 000	blue-white
A	10 000–7500	white
F	7500–6000	yellow-white
G	6000–5200	yellow
K	5200–3700	orange
M	3700–2000	red

PHYSICS ONLY

The brightness of a star



▲ Figure 27.5 It is impossible to decide which of these vehicles has the brightest headlights just by looking at the photograph.

At first there would seem to be little doubt which of these vehicle headlights is the brightest. But is it really that simple? The cars in the distance might have brighter headlights but they appear to be dimmer because they are further away. Also the brightness of headlights in different makes of cars and lorries could be different!

When we look at the stars in the night sky we have similar problems deciding which are the brightest.

The brightness of a star depends on **a** the distance the star is from the Earth and **b** what the star is made from and the kinds of nuclear reactions that are taking place.

As a result there are three different ways in which astronomers describe the brightness of a star:

1. The apparent brightness or magnitude of a star. This is the easiest method and is simply a measure of how bright a star is as seen from the Earth.
2. The absolute brightness or magnitude. This is a measure of how bright stars would appear if they were all placed the same distance away from the Earth. (This distance that astronomers agreed on, called the standard distance, is equal to 10 parsecs or 32.6 light years). This is often a better measure of brightness because it allows scientists to make comparisons between stars.
3. The luminosity of a star. This measures how much energy in the form of light is emitted from a star's surface every second.

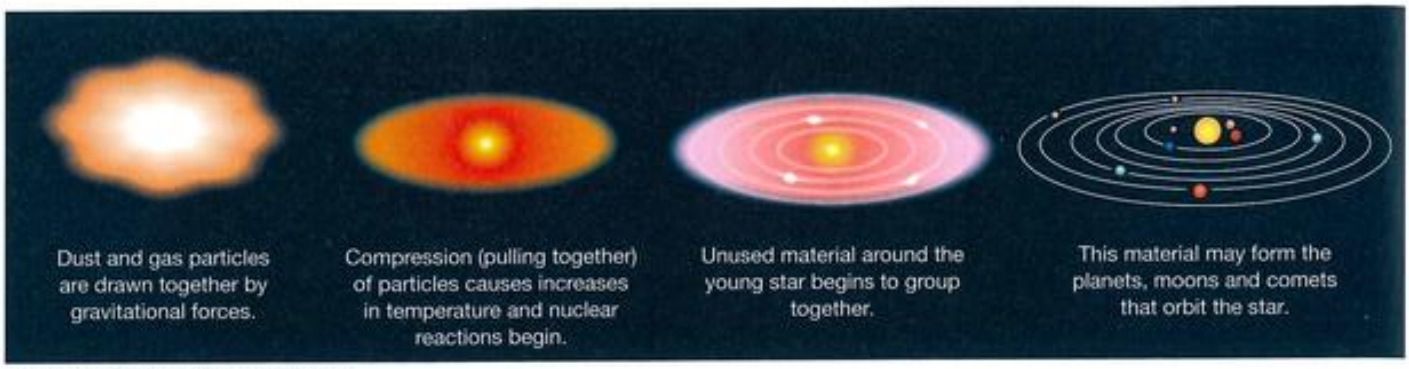
END OF PHYSICS ONLY

REMINDER

The brightness of a star depends on:

- the distance the star is from the Earth
- the luminosity of the star – that is, how much energy is transferred from the star each second.

THE BIRTH OF A STAR



▲ Figure 27.6 Birth of a solar system

Stars like our Sun are formed from large clouds of dust and gas particles we call nebulae or stellar nebulae. These particles are drawn together over a very long period of time by gravitational forces. The particles are pulled together so tightly that there is a very large increase in temperature and pressure. As a result of this nuclear fusion reactions begin. Hydrogen nuclei join together to make larger nuclei and huge amounts of energy in the form of heat and light are released. This incredibly hot ball of gas is a very young star.

THE LIFE OF A STAR

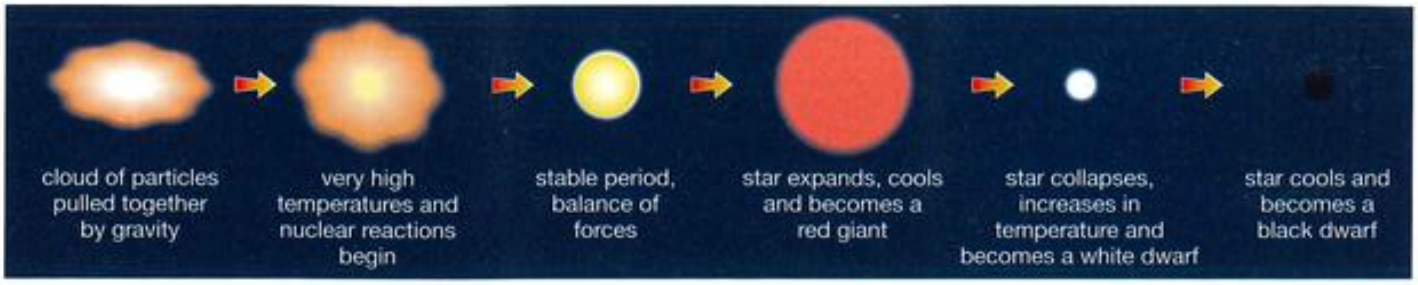
The appearance of a star changes gradually with time. These changes follow a pattern.

When a star first forms, gravitational forces are pulling particles together. Then when nuclear reactions begin, the high temperatures create forces that try to push the particles apart – that is, make the gases expand. When these two forces are balanced, the star is said to be in its main stable period. A star in this main stable period is referred to as a main sequence star (you will learn why later). This period can last for many millions of years. At the moment our Sun is in this stable period.

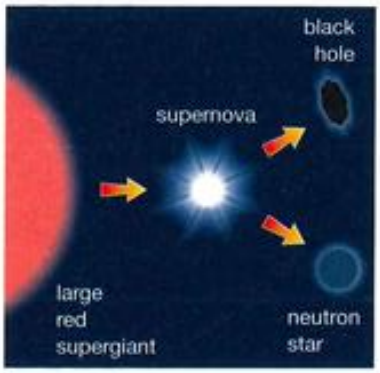
Towards the end of this stable period, there are less hydrogen nuclei and eventually the hydrogen fusion reactions stop. Gravitational forces are now the largest forces and compress the star. As the star shrinks in size there is a large increase in temperature. So high that fusion reactions between helium nuclei begin. The energy released by these reactions causes the star to expand to many times its original size. As it expands it becomes a little cooler and more of its light energy is emitted in the red part of the spectrum. The star is changing into a red giant.

Sometime later when most of the helium nuclei have fused (joined) together, new nuclear reactions begin, but now the compressive or squashing forces are larger and the star begins to get smaller or contract. This contraction causes an increase in temperature so the star again changes colour. It now emits more blue and white light. It has changed into a white dwarf star. The matter from which white dwarf stars are made is millions of times more dense than any matter found on the Earth.

Finally as a white dwarf star cools it changes into a cold black dwarf star.



▲ Figure 27.7 The life-cycle of stars with a mass similar to that of our Sun



▲ Figure 27.8 The final stages for stars with masses much greater than our Sun

A star that is much larger than our Sun may follow a slightly different path at the end of its life. After the stable period it will expand into a large red supergiant. At the end of this period, as it contracts it becomes unstable. It explodes throwing dust and gas into space to form a new stellar nebula. This exploding star is called a supernova. Any matter remaining will form a very dense neutron star. If the neutron star has a mass that is approximately five times greater than that of our Sun or more, it collapses further to become a black hole.

EXTENSION WORK

The gravitational field of a black hole is so strong that even light is unable to escape from it. This is why they are called 'black holes'.
All the elements in the periodic table up to iron are formed in stars but elements heavier than these are only formed in supernova explosions.

PHYSICS ONLY

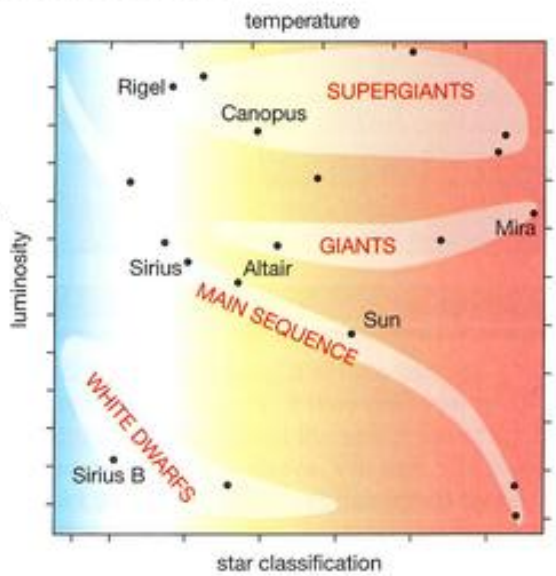
The Hertzsprung–Russell diagram (HR diagram)

Two scientists, Ejnar Hertzsprung and Henry Norris Russell, separately discovered the relationship between the brightness of a star and its surface temperature. They created a diagram that shows the relationship between the brightness, temperature and classification of a star.

The importance of this diagram is that by finding a star's position in the diagram we then know its internal structure and at what stage it is at in its life.

For example, Sirius lies in the main sequence belt so it is a star similar to our Sun and fusion reactions between hydrogen nuclei are taking place in its core. In the next stage of its life it will become a red giant.

Rigel is in the supergiants region of the diagram and so has already passed through the main sequence part of its life. The majority of fusion reactions taking place in its core are between helium nuclei. It will eventually become a neutron star or a black hole.



▲ Figure 27.9 The Hertzsprung–Russell diagram is important because it can be used to describe the life story of a star and to predict the final stages of its life.

HINT

Try to remember the general features of the HR diagram – that is, the shape and position of the main sequence and the positions of white dwarfs, red giants and red supergiants.

END OF PHYSICS ONLY